A nuclear-register criterion for radical-pair computation in cryptochrome

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Spin coherence lets a radical pair in cryptochrome sense the geomagnetic field, and it has long been conjectured that the same chemistry could support biological computation. Yet the conjecture has never been put in a form that chemistry can test: no proposal names the quantity a downstream reaction would read, the clock on which it would be read, or the relaxation constant that bounds it. Here we derive that statement—a *register-reuse criterion*—by treating a cryptochrome-type radical pair as a reservoir computer, solving its master equation with electronic decoherence and driving it as a real turnover does, by the birth of a spin-correlated pair. We find that the memory is nuclear, not electronic: the register survives recombination in the diamagnetic product, so the clock is the turnover interval rather than the microsecond pair lifetime, and the computation is readable by ordinary chemistry through the polarisation the product carries away (out-of-sample capacity 4.4 against a memoryless floor of 2.0), provided the *same* nuclear register is reused between turnovers. We then predict that register's relaxation instead of assuming it. At the geomagnetic field a protein-bound proton relaxes in milliseconds, not seconds, and the register the biology can keep—the tryptophan proton alone, carried by the product—caps the memory horizon at 1.7–2.6 ms, and at about 1 ms once the molecule budget of the cycle is closed, below the neural band unless its dipolar vector reorients four to six (closed: ten to fifteen) times faster than the protein ($\tau_{\mathrm{eff}} \lesssim 2.6$–$4.0$ ns; closed, 1.0–1.6 ns). Three further results bound the criterion: a classical network matched channel-for-channel performs as well as the five-spin reservoir once both readouts are regularised by cross-validation, so no quantum advantage can be invoked; the readout is counting-noise limited below about 10^9 molecules per pool; and the drive cannot be light. What the conjecture reduces to is a criterion whose constants—the turnover interval, the product's proton T_1 at 50 μT and the pool size—can be measured in vitro, by steady-state kinetics and relaxation dispersion, before any neuron is involved.

I. INTRODUCTION

The radical-pair mechanism (RPM) is the only experimentally established way in which a magnetic field as weak as the geomagnetic field (~ 50 μT) can change the outcome of a chemical reaction[1,2,16]. A spin-correlated pair born in a singlet electronic state undergoes coherent singlet–triplet (S–T) interconversion driven by hyperfine fields; because recombination is spin-selective, a field that re-tunes this interconversion alters the product yields. The mechanism underlies the leading model of avian magnetoreception, where the photoreceptor cryptochrome hosts a flavin–tryptophan radical pair[2–5,12].

The success of the RPM has repeatedly motivated a bolder conjecture: that spin coherence in biomolecules might support not merely sensing but *information processing*—a "quantum brain"[6,7], a proposal long contested on decoherence grounds[9] and part of the broader programme of quantum biology[10,11]. A concrete, named instance is the *three-layer quantum-brain hypothesis*[19,20], which posits a layered architecture—a long-lived nuclear-spin memory (Layer 1), a radical-pair read/write interface (Layer 2), and classical neuronal transduction (Layer 3)—and which, in its most recent form, already reframes the proposal from quantum *computation* to quantum *reservoir dynamics*. The present paper is the quantitative test of that architecture as a reservoir, and the bound it returns. Since that reframing is not new here, we state the delta explicitly: Ref.[20] proposes the reservoir reading and estimates coherence lifetimes, but drives the reservoir by resetting a single electron, reads it through spin observables that presuppose magnetic-resonance access, and identifies the memory horizon with the radical-pair lifetime. What is new here is (i) a chemically consistent drive in which the electron pair is reborn correlated each turnover, (ii) readouts restricted to quantities a downstream reaction could use, each scored against its own memoryless floor, (iii) classical baselines matched by readout dimension, and (iv) the replacement of the lifetime clock by the turnover clock together with the register criterion that follows from it. The spin Hamiltonian, the Haberkorn treatment and the capacity estimators are standard and are not claimed as contributions. Only a *spatially separated* radical pair, as in cryptochrome—with a microsecond lifetime and an electron–electron coupling weak enough not to lock the pair in one spin state—sustains the coherence required for weak-field sensitivity[23]. For the FAD–W324 pair at about 2 nm the exchange and dipolar couplings are each of order 10 MHz and partially cancel[23]; this premise is a standard result of radical-pair compass theory[2,23], and we take the existence of a long-coherence separated radical pair as our self-contained starting point.

That premise invites the question we address here. The separated radical pair is singled out because it preserves

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long-lived electron–nuclear spin coherence. *Is that coherence good for anything besides sensing the geomagnetic field?* We argue the question is best posed not as gate-model quantum computation but as *reservoir* computing, and we answer it with the chemistry left in: driven, read and clocked as a turnover actually is. We state at the outset the load-bearing caveat: whether a *neuronal* cryptochrome forms a light- or reaction-driven separated radical pair at all is unestablished—the mammalian CRY1/CRY2 proteins are primarily circadian transcriptional regulators[15]—so every biological conclusion below is conditional on the existence of such a pair. The spin-dynamics results (capacity, classical baselines, the register criterion) do not depend on that condition and stand on their own.

Why a reservoir, not a gate. Gate-model quantum computation treats decoherence as the adversary and demands error correction; a warm, wet, 10^{10}-copy biomolecular ensemble cannot meet that bar. Reservoir computing (RC) inverts the relationship. An RC system feeds a time-dependent input into a fixed nonlinear dynamical "reservoir" and trains only a *linear* readout of its state[24,25]. RC *requires* dissipation: the echo-state / fading-memory property—that the influence of past inputs decays—is supplied precisely by relaxation[26,27]. The molecular-spin and NMR communities have demonstrated quantum RC explicitly in nuclear- and electron-spin ensembles[26–28], part of a broad physical-reservoir-computing effort[13]. That field has moved in the last two years in directions that bear directly on the present question. Dissipation is now treated explicitly as a computational resource rather than a nuisance[42]; memory longer than the coherence time has been obtained by measuring mid-circuit, resetting, and feeding the classical record back in[43,44], which is to say by keeping the memory somewhere other than the coherent state; whether entanglement helps at all is found to depend on how fast the input fluctuates relative to the dissipation[45]; and a nine-spin correlated-spin reservoir read out with full quantum access has been reported to outperform classical reservoirs of thousands of nodes on forecasting benchmarks[46]. Our setting differs from all of these in the one respect that decides the outcome below: the readout is restricted to what chemistry can carry away, and the memory is carried by a nuclear register retained between cycles, not by the coherent state. In this system the fading memory is supplied not by electronic dephasing—varying T_2^e over the physically plausible range 50–2000 ns at the engineered operating point changes the capacity by under 2% (the two operating points are defined in Methods and must not be conflated)—but by the turnover itself: each cycle discards the electron pair and keeps only the nuclei, which is a dissipative, non-unitary map. It is the openness of the chemical cycle, not the rate of electronic decoherence, that makes the system a reservoir. This is consistent with the large coherence fraction reported below, which is measured in a very different regime: dephasing rates 10^3 times larger than any realistic T_2^e, applied together with a compensating extension of the pair lifetime. Realistic electronic decoherence is not the limiting factor; complete destruction of electronic coherence still is.

Here we answer that question with the chemistry left in, and the answer is a criterion rather than a verdict. We drive the reservoir as a real turnover does—nuclei retained, a new spin-correlated pair created each cycle—rather than by resetting a single electron, and read it only through the products a downstream reaction could use. We find that the memory is carried by the nuclei of the diamagnetic product, so the reservoir's clock is the turnover interval and not the radical-pair lifetime on which the usual timescale objection rests. The constraint that replaces that objection is the *register-reuse criterion*: the same nuclear register must be reused from one turnover to the next, and it must survive the interval. We then make the criterion quantitative by predicting the register's relaxation at the geomagnetic field from measured couplings, which fixes the two constants an experiment would have to measure. Two further results bound what the criterion can deliver. Classical reservoirs matched by readout dimension exceed the spin reservoir, so there is no quantum advantage to invoke, and the drive cannot be photochemical. The criterion still compares two times, as the argument it replaces did; what the spin-dynamical treatment supplies, and no timescale comparison would, is which nuclear relaxation constant appears in it, and how large that constant is.

The physical correspondence we adopt, adapting the spin-RC framework of Fujii and Nakajima[26], maps the reservoir-computing elements onto the radical pair as follows: the memory nodes are the nuclear spins (in the cryptochrome case ^{14}N and ^{1}H), which store the input in their *populations*, so that the relevant relaxation constant is T_{1n}—though at the geomagnetic field extreme narrowing makes T_{1n} and T_{2n} equal, and both short (supplementary material, Sec. S12); the nonlinear read/write head is the radical-pair electron system; the read/write channel is the hyperfine coupling; and the fading memory (echo-state property) is supplied by the openness of the chemical cycle together with inter-cycle nuclear *depolarisation*—not nuclear dephasing, which we show below costs the reservoir almost nothing. These are exactly the three layers of the hypothesis above: Layer 1 (nuclear-spin memory), Layer 2 (the radical-pair interface), and Layer 3 (the classical readout/transduction that trains the linear map).

The paper is organized as follows. Section II establishes the reservoir's capacity and its convergence, localises that capacity in spin coherence, compares it against matched classical baselines, and then asks what a chemical readout recovers; it closes with the reservoir's clock and the nuclear-register criterion. Section III places the result among quantum-brain proposals and states the limitations. The model, protocol, and capacity definitions are collected in Appendix A.

II. RESULTS

We drive the reservoir the way a real turnover does. Each cycle the nuclei are retained and a *new* spin-correlated electron pair is created whose singlet character encodes the input, $\rho_e(s) = s\,|S\rangle\langle S| + (1-s)P_T/3$; the pair then evolves under the Liouville-space master equation with electronic dephasing and, where the readout requires it, spin-selective recombination. This replaces the reset of a single electron used in earlier treatments, which is not a chemical process and which—for a separated pair—destroys the electron–electron correlation every cycle. All capacities are out-of-sample (trained on the first half of a run, scored on the held-out second half). The number of input realisations is not uniform and is stated with each result: 12 for the turnover-clock scan and its paired test, which carry the central claim; 8 for the classical baselines and the T_2^e scan at the engineered point; 6 for the headline capacity, the chemical-readout survey, the route floors, the delay kernel, the grid scans, the nuclide scan and the cryptochrome-point classical baselines; 4 for the register-reuse and nuclear-channel scans and the semiclassical reference; and 3 for the ensemble and anisotropy scans. Four quantities are single-realisation: the ridge and shot-noise sensitivity scans and the semiclassical integration-step check, which compare variants on one fixed input by design, and a single-realisation clock scan retained as supplementary material, Table S11, whose single realisation is why it is a control and not a result. Model, protocol and estimator are given in Appendix A.

A. Capacity of the reservoir

With a twelve-observable readout the five-spin reservoir attains

$$\mathrm{MC} = 3.9, \qquad \mathrm{IPC} = 5.6 \pm 0.2,$$

of which 1.7 units are nonlinear. Unlike the single-electron-reset protocol, the estimate is converged in the length of the driving sequence: repeating at $L = 600, 1200, 2400, 4800$ gives IPC $= 5.28, 5.60, 5.87, 5.84$, i.e. flat to within the seed-to-seed scatter beyond $L \approx 2400$. The memory kernel decays over ~ 4 cycles, the signature of the fading-memory property.

B. An operational coherence fraction

To ask how much of this is quantum we add a pure-dephasing rate γ in the S_z product basis and take the incoherent limit at *matched hopping rate*: since the golden-rule transfer rate scales as $|H_{ij}|^2/\gamma$, holding rate $\times\,\tau$ fixed requires $\tau \propto \gamma$. Doing so, the capacity saturates at IPC $= 1.00$ for $\gamma = 10^3, 5\times10^3, 2\times10^4$ MHz — exactly the memoryless floor, i.e. instantaneous read-back

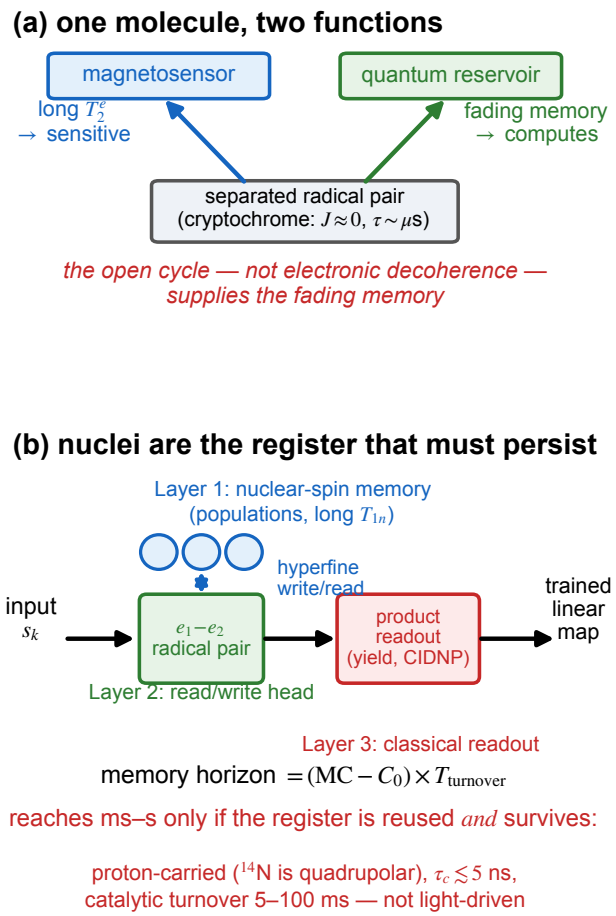


FIG. 1. **The reservoir and the quantity that decides its reach.** (a) A cryptochrome-type separated radical pair (weak electron–electron coupling, lifetime $\tau \sim \mu$s) is the same molecule in two roles: long electronic coherence makes it a weak-field magnetosensor, while the dissipation of an open cycle supplies the fading memory a reservoir needs. (b) The nuclei are the register: they are not consumed by recombination and therefore persist from one turnover to the next, whereas the electron pair is created afresh each cycle. The memory horizon is consequently set by T_{turnover} and not by τ, and is quoted throughout as $(\mathrm{MC} - C_0) \times T_{\mathrm{turnover}}$, the zero-delay term $C_0 \simeq 1$ carrying no memory. That quantity is a delay capacity integrated over past turnovers, not the longest delay that can be reconstructed; the two agree here to within a factor of two because the delay profile is monotone (linear capacity 0.85, 0.47, 0.18, 0.04 at one to four turnovers back, supplementary material, Sec. S2), so no horizon quoted here hides a long tail. It reaches the neural band only if the register is genuinely reused *and* survives the pause, which requires it to be carried by protons rather than the quadrupolar flavin ^{14}N and to reorient with $\tau_c \lesssim 2.6$–4.0 ns (supplementary material, Secs. S9 and S12).

of the input with no memory at all. Against the coherent value 5.42 this is a coherence fraction

$$\frac{\mathrm{IPC}_{\gamma=0} - \mathrm{IPC}_{\mathrm{incoh}}}{\mathrm{IPC}_{\gamma=0}} = 0.82. \quad (1)$$

Taking the limit at fixed τ instead, as is sometimes done, measures Zeno freezing rather than the classical limit and inflates this number; we avoid that.

Two caveats are load-bearing. First, γ as applied above acts on all five spins, so one might suspect the collapse merely reflects erasure of the nuclear register rather than loss of electronic coherence. It does not: restricting the added dephasing to the two electrons and leaving the nuclei untouched gives IPC = 0.9949, 0.9891 and 0.9891 at the same three rates, i.e. a coherence fraction of 0.817 against 0.816 for the all-spin channel (supplementary material, Sec. S5). The number is a statement about the electrons. Second, and more limiting, this is an *operational* fraction defined by one particular matched-hopping construction in which τ is stretched by up to a factor of 400; it is not a decomposition against a physically motivated classical model. The appropriate such reference is a semiclassical radical-pair treatment with classical hyperfine vectors and rate equations[16]. We run it here (supplementary material, Sec. S8): a fully classical model of the same spin system scores at its own memoryless floor, 0.76 against a floor of 0.77 at 512 trajectories, so it holds no memory the quantum system does not, but its absolute capacity is limited by trajectory sampling rather than by physics, and we do not quote the coherence fraction it would imply as a converged number. We report 0.82 as what it is—the capacity lost under a specific incoherent limit—rather than as a measure of "how quantum" the reservoir is. Note also that this quantity, like the capacity above it, is evaluated at the engineered operating point.

C. There is no quantum advantage

A capacity is only meaningful against a classical reference, and the comparison must be symmetric in what is read out. Driving tanh echo-state networks (ESNs) with the same input and scoring them with the *same* estimator:

TABLE I. The quantum reservoir against matched classical reservoirs at the engineered operating point.

reservoir (readout features)	IPC
quantum, 5 spins (12 observables)	5.59 ± 0.19
ESN, 5 nodes (5 linear features)	4.55 ± 0.32
ESN, 5 nodes (5 linear + 7 quadratic)	9.41 ± 0.86
ESN, 12 nodes (12 linear features)	10.37 ± 0.59

Matched by *physical unit* the quantum reservoir is modestly ahead (5.59 vs 4.55), but matched by *readout dimension*—the resource that actually sets the Dambre ceiling—a classical system of the same size is well ahead (9.41 vs 5.59). We therefore make no claim of quantum advantage. Which accounting is the fair one depends on a cost model for non-commuting measurements versus multiplications that we do not have; we report the numbers and leave the claim unmade.

That comparison is made at the engineered operating point, which is not the biologically relevant one. Repeating it at the cryptochrome point, against ESNs matched channel-for-channel to the chemical readouts of the next section, removes the ambiguity entirely:

TABLE II. The same comparison at the cryptochrome operating point, matched channel-for-channel. Left: the ridge fixed at $\lambda = 10^{-6}$ on raw features, as everywhere else in this paper (six seeds). Right: features standardised on the training half and λ chosen per target by five-fold cross-validation on that half, for both systems (twelve paired seeds).

system (readout features)	λ fixed	λ by CV
quantum, kinetics (5 ch)	1.97 ± 0.10	4.64 ± 0.29
ESN, 5 nodes (5 features)	4.47 ± 0.33	4.55 ± 0.30
quantum, +CIDNP (8 ch)	4.42 ± 0.18	6.90 ± 0.47
ESN, 8 nodes (8 features)	6.94 ± 0.56	6.95 ± 0.61
ESN, 12 nodes (12 features)	9.62 ± 0.93	—

At the cryptochrome point, with the ridge fixed at $\lambda = 10^{-6}$ on raw features, the classical reservoir wins whenever the comparison is matched channel-for-channel: by 57% at eight channels and 127% at five. That margin is not a property of the two systems but of the regulariser. The chemical channels have amplitudes of order 10^{-2} and the ESN states of order one, so one absolute ridge penalises the quantum readout far more heavily (Sec. III). Standardising the features and letting each target choose its own λ by cross-validation—for both systems, on the training half only—removes the margin entirely: the paired difference is $+0.06 \pm 0.16$ (SEM, $n = 12$) at eight channels and -0.10 ± 0.10 at five, that is +1% and −2% of the quantum value, neither resolved from zero. Matched by *physical unit* instead—five spins against five nodes—the fixed-λ numbers are 4.42 ± 0.18 against 4.47 ± 0.33, again indistinguishable. The honest summary is therefore not that the classical system wins but that the two are equivalent at matched channel count: on no accounting does the radical pair compute *better* than a small classical network by a margin exceeding its own seed scatter, and on no properly regularised accounting does it compute worse. The quantum advantage is absent; so is a quantum deficit. It is also the opposite of what a correlated-spin reservoir with full quantum readout has recently been reported to achieve[46]; the two results are not in tension, because what is read out differs. That experiment trains on the full set of spin observables of an engineered many-body system; here the readout is the product yield and the polarisation it carries, and a

classical network given the same number of channels does as well. The advantage, where it exists, lives in the readout access, which is exactly what a chemical cycle does not grant.

D. Biology can read the reservoir

Biology has no EPR spectrometer; it has products. We therefore restrict the readout to observables a downstream reaction could plausibly use, with recombination switched on so that the quantity read is the yield actually produced, $Y_X = k_X \int_0^\tau \mathrm{Tr}[P_X \rho(t)]\, dt$ (Fig. 2):

The memoryless floor is *not* common to these routes, and quoting a single floor of IPC = 1 overstates what the richer readouts achieve. We therefore measure each route's own floor by wiping the nuclear register every cycle ($q = 1$), which leaves only what the route can compute from the present input, and report the excess above it. All seven routes defined by the model are listed; none is omitted.

Four things follow, and only the first is the expected one. First, the readout is genuinely not the limiting resource: time-resolved product kinetics clear their own floor by a factor of two, and the nuclear polarisation the product carries away reaches 4.42 against a floor of 2.00—an excess of 2.42, not the factor of 4.4 that a universal floor of 1 would suggest. The CIDNP route is still the best readout by a wide margin, and it still requires no spectroscopic access.

Second, a single scalar end-point yield gives IPC = 1.02 against a floor of 1.02: it reports the present input and remembers *nothing*. Claims about reservoir behaviour should not be based on it.

Third, the readout saturates in channel count. Doubling the time-resolved kinetics from 5 to 10 channels by adding the second exit channel buys 0.003 in capacity. Whatever limits this reservoir, it is not the number of product observables a downstream reaction can distinguish.

Fourth—and this is a caution about the accumulated-pool route rather than a result—a downstream integrator carries memory of its own. Its floor is 1.96 with a linear memory capacity of 1.94 even when the nuclear register is wiped every single cycle, because the leaky accumulator is itself a classical delay line. Its apparent capacity is therefore not evidence of nuclear-spin memory, and we exclude it from the claims below.

The mechanism behind the CIDNP column is worth stating, because it also explains a negative result. The reduced state of *either* electron of a newly born pair is maximally mixed for every input, since both $|S\rangle\langle S|$ and $P_T/3$ have zero one-electron polarisation; the input lives entirely in the two-electron correlation. Under the reference premise ($J = 0$, no dipolar term) the two radicals do not interact, so nothing but recombination can transfer that correlation to a nucleus—and indeed, with recombination switched off, the cryptochrome-point capacity is 1.02, the memoryless floor. It is *spin-selective recombination* that converts the singlet–triplet correlation into a population difference and so writes the input into the nuclear register. That exclusivity is a property of the premise, not of the pair: with the electron–electron coupling the source for the separated pair actually reports[23], the coupled electrons transfer the correlation without recombining, and the same no-recombination capacity is 5.1–5.9 (Sec. II E). What is premise-independent is the mechanism at the product: recombination writes the input into the register whether or not it is the only thing that can.

E. What depends on the coupling premise

The cryptochrome point idealises "separated" as $J = 0$ with no dipolar term, so that the radicals interact only through recombination. The source for the separated pair reports otherwise[23]: for FAD–W324 at $r = 1.90$ nm, $D = -11.4$ MHz and an exchange gap of 1.3–126 MHz of undetermined sign, partially cancelling. We therefore rerun every cryptochrome-point quantity with those values in Eq. (A1) and nothing else changed (supplementary material, Sec. S14), and state what moves. With recombination switched off, the capacity rises from the memoryless floor, 1.02, to 5.1–5.9 across both signs of J and both orientations of the inter-radical axis; even 0.6 MHz of D, the value at $r = 5$ nm, gives 5.2. The reference premise is thus the one point of the coupling range at which recombination is the *only* write mechanism. With recombination on, the chemically accessible routes move as Table S31 records: the time-resolved kinetic capacity moves in both directions, from 2.0 to 1.3–3.4, and so does CIDNP, 3.9–5.2 against 4.42, with every floor unchanged. The largest quantum-side value in that range, 3.4 at five channels and 5.2 at eight, remains below the matched classical network, and the kinetic route still falls to its floor when the register is restricted to the proton. What does not move is what the criterion rests on. The end-point yield stays on its floor, the floors themselves do not shift, and the two constants the criterion compares—the nuclear T_1 and the turnover interval—are untouched by the coupling. The ceiling they imply does move, because the coupled electrons write more polarisation per cycle: computed with the same eight-point scan for both premises (which gives 3.4 ms for the reference premise where the refined scan of Sec. II G gives 3.5), it rises from 3.4 to 5.7 ms with the intermolecular bath, still below the band, and the correlation time at which the band opens moves from 5.2 to 8.7 ns, so the register must reorient 1.8 rather than 2.9 times faster than the protein. The criterion survives the premise; its margin is what the premise sets.

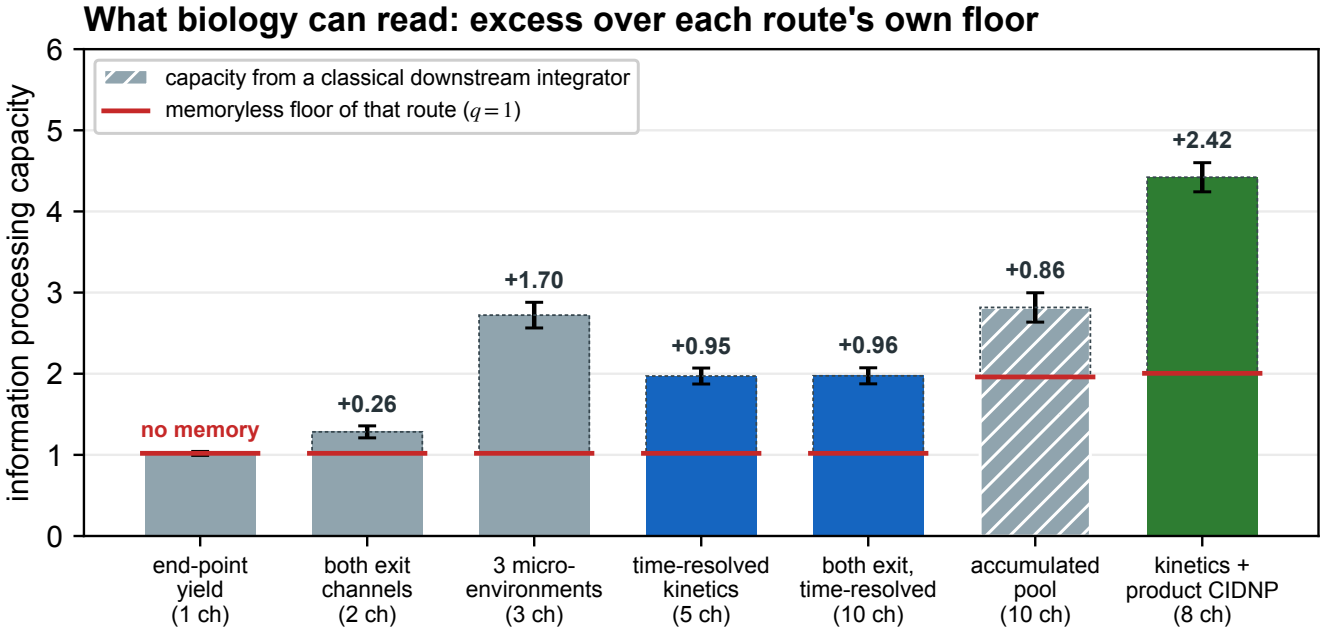


FIG. 2. **Readout is not the bottleneck.** Capacity recovered by chemically accessible observables at the cryptochrome operating point ($B = 50$ μT, $J = 0$, $\tau = 1$ μs), with spin-selective recombination switched on so that the observable is the product actually formed. All seven routes defined by the model are shown. Red bars mark each route's *own* memoryless floor, measured by wiping the nuclear register every cycle ($q = 1$); these are not common to the routes (1.02 for the yield routes, 2.00 for the eight-channel CIDNP route, 1.96 for the accumulated pool), so it is the dashed excess above each floor—printed above each bar—that measures memory, not the bar height. Ordinary reaction kinetics clear their floor by 0.95, and the nuclear polarisation carried away by the product—chemically induced dynamic nuclear polarisation, CIDNP[18]—clears its own by 2.42, the largest of any route. The end-point yield sits exactly on its floor: it reports the present input and remembers nothing. The accumulated pool is hatched because a leaky downstream integrator is itself a classical delay line, so its excess is not evidence of nuclear-spin memory. No spectroscopy is involved.

TABLE III. Capacity recovered by chemically accessible observables at the cryptochrome point, each scored against its own memoryless floor.

readout route	ch.	IPC	floor	excess
end-point singlet yield	1	1.02	1.02	0.00
both exit channels, end point	2	1.28	1.02	0.26
three micro-environments (different τ)	3	2.72	1.02	1.70
time-resolved kinetics, one product pool	5	1.97	1.02	0.95
both exit channels, time-resolved	10	1.97	1.02	0.96
kinetics + CIDNP on the product	8	**4.42**	2.00	**2.42**
kinetics + accumulated downstream pool	10	2.82	1.96	0.86

F. The clock is the turnover, not the pair lifetime

The horizon over which a reservoir remembers is MC multiplied by its cycle time. It is natural to identify that cycle time with the radical-pair lifetime $\tau \sim 1$ μs and conclude that a 3 μs horizon puts the system 10^3–10^5 below neural timescales. That identification is wrong, and its own model says so. The carrier of the memory is the nuclear register, and nuclei are not consumed by recombination: they leave in the diamagnetic product, where nuclear relaxation times are of order 0.1–1 s—a *high-field* figure, which Sec. II G and the The supplementary material, Sec. S12, revises sharply downwards for the geomagnetic field. The cycle time of the reservoir is therefore the interval between turnovers.

Inserting a pause T_d between turnovers and relaxing the register with probability $q = 1 - \exp(-T_\mathrm{d}/T_{1n})$ (with $T_{1n} = 1$ s) gives Fig. 3(a):

Before the numbers, the horizon needs a definition that is actually a time. MC is a sum of squared correlations over delays $d \geq 0$, and the $d = 0$ term—which reports

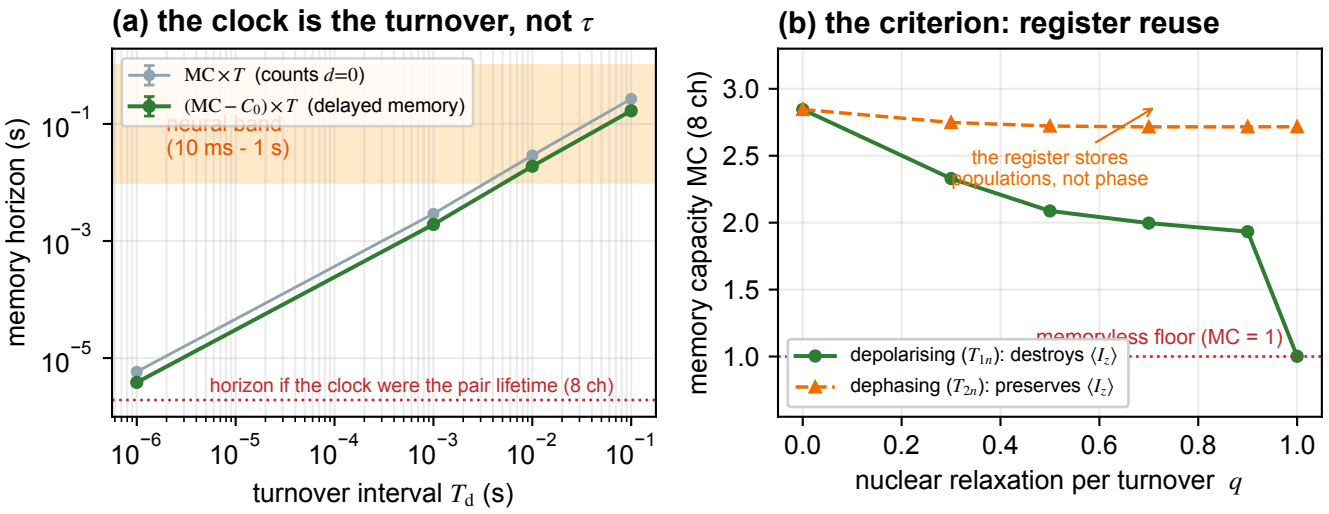


FIG. 3. **(a)** The memory horizon against the turnover interval, from twelve input realisations (error bars are the between-seed standard deviation). Because the nuclear register survives recombination, inserting a physiological pause costs only 8% of the memory capacity across five decades, so the horizon simply tracks the clock and enters the neural band (shaded, 10 ms–1 s, the range of cortical rhythms[8]). Both definitions are shown: the raw MC × T_{turnover} (grey) counts the zero-delay term, which carries no memory, while the delayed-memory horizon $(\mathrm{MC} - C_0) \times T_{\mathrm{turnover}}$ (green) is the quantity quoted in the text—on the latter the 1 ms turnover falls below the band. The dotted line is where the horizon would sit if the clock were the radical-pair lifetime, for the same eight-channel readout. **(b)** The criterion, and which relaxation process enforces it. Depolarising the register between turnovers with probability q (green) drives the capacity to the memoryless floor: at $q = 1$—a fresh molecule each cycle—MC = 1.00 and there is no memory. Pure nuclear *dephasing* at the same q (orange) leaves MC = 2.72, a cost of under five per cent: the register stores populations, not phase, so the constant that bounds the horizon is T_{1n}. Note that this does not lengthen the horizon at the operating point, where extreme narrowing brings T_{1n} and T_{2n} together (supplementary material, Sec. S12).

the present input and remembers nothing—contributes $C[0] \simeq 1$ for every readout. Multiplying the raw MC by the cycle time therefore assigns a positive "horizon" to a system with no memory at all, growing without bound as the cycle is lengthened. We instead quote $(\mathrm{MC} - C[0]) \times T_{\mathrm{cycle}}$, the delayed memory only. The measured kernel justifies this: for the CIDNP readout $C[0] = 1.00$, $C[1] = 0.85$, $C[2] = 0.47$, falling below 5% of $C[0]$ by $d = 4$, so $\mathrm{MC} - C[0] = 1.82$ of genuinely delayed capacity.

Twelve input realisations; the quoted spread is the between-seed standard deviation. Because the same seeds are used at every T_{d}, the change is properly assessed on paired differences: across five decades of turnover interval the capacity falls by 0.239 ± 0.028 (SEM, $n = 12$, $t = 8.46$), that is by 8.2%. This is a real decrease, resolved well above both the seed noise and the time-grid error (0.03% from $n_t = 96$ to 192), and it is concentrated at the longest pause: the first three decades cost under one per cent and the last costs seven. A single-realisation scan of the same quantity, kept as a control in the supplementary material (Sec. S7), cannot resolve a change of this size, which is why twelve realisations are used here. The conclusion rests on that sample: a 10^5-fold change in the clock moves the capacity by under a tenth, so the horizon tracks the turnover interval and not the pair lifetime.

A turnover of 10–100 ms therefore places the memory horizon at 19–168 ms, inside the neural band, *provided the register survives the pause*. That proviso is not free, and Sec. S12 shows it is what now binds: at the geomagnetic field a proton on the intact protein relaxes in 2.4 ms, which caps the horizon at 5.4 ms on the intramolecular term alone (3.5 ms with the intermolecular bath) and puts it below the band unless the register reorients 1.8 (bath: 2.8) times faster than the protein—and that is with all three nuclei assumed to persist; on the register the biology can actually keep, the cap is 2.6 (1.7) ms and the requirement 3.8 (5.9) times (Sec. II G). The numbers in this table should therefore be read as what the spin dynamics permit, not as what the chemistry delivers. We note also that a 1 ms turnover gives 1.9 ms, *below* the 10 ms–1 s band drawn in Fig. 3(a), so the band is reached only at turnovers of 10 ms and longer. The timescale objection based on the microsecond pair lifetime does not survive in any of these cases: the same spin physics runs on a slow clock.

G. The criterion: reuse of the nuclear register

What replaces it is a chemical requirement, and a sharper one. Everything above assumes that the reg-

TABLE IV. The memory horizon against the turnover interval, twelve input realisations.

T_d	q	MC (8 ch)	MC $\times\, T$	(MC $- C_0$) $\times\, T$
1 μs	10^{-6}	2.919 ± 0.205	5.8 μs	3.8 μs
1 ms	10^{-3}	2.916 ± 0.204	2.9 ms	1.9 ms
10 ms	10^{-2}	2.894 ± 0.196	29 ms	19 ms
100 ms	9.5×10^{-2}	2.679 ± 0.124	268 ms	168 ms

ister the reservoir writes into is the *same* register on the next turnover. Relaxing that assumption by depolarising the nuclei with probability q between cycles gives Fig. 3(b): the eight-channel memory capacity falls from 2.85 at $q = 0$ to 2.33 at $q = 0.3$ and 2.09 at $q = 0.5$—a loss of more than a quarter by the time the turnover interval reaches the nuclear relaxation time—then through 2.00 at $q = 0.7$ and 1.93 at $q = 0.9$, and at $q = 1$, a fresh molecule every cycle, collapses to MC = 1.00, the memoryless floor.

Which nuclear relaxation process enforces this can be asked directly. The register we propagate is a full nuclear density matrix, so one might expect memory to require the survival of nuclear coherences as well as populations, making the binding constant $\min(T_{1n}, T_{2n})$. It does not. Replacing the depolarising channel by a pure-dephasing channel on each nucleus, which destroys nuclear coherence while preserving $\langle I_z \rangle$ exactly, leaves the capacity within five per cent: MC = 2.85 at $q = 0$ and 2.72 at $q = 1$, against 1.00 for the depolarising channel at $q = 1$.

TABLE V. Which nuclear relaxation process enforces the criterion.

q	MC (8 ch), depolarising	MC (8 ch), dephasing
0.0	2.846	2.846
0.5	2.088	2.722
0.9	1.933	2.716
1.0	1.002	2.716

The memory is carried entirely by nuclear z-populations—which is exactly what the CIDNP readout reports—and is indifferent to nuclear phase. The relevant constant is therefore T_{1n} and not T_{2n}.

We must immediately withdraw the rescue this appears to offer. At the geomagnetic field the register is in extreme narrowing ($\omega\tau_c \approx 2 \times 10^{-4}$), and there $J(0)$, $J(\omega)$ and $J(2\omega)$ coincide, so T_{1n} and T_{2n} agree to one part in 10^7 (supplementary material, Sec. S12). The distinction between a population register and a coherence register is therefore real as a statement about what the reservoir uses, but it buys nothing at the operating point: both constants take the same, short, value there. It would matter only at fields where $\omega\tau_c \gg 1$, that is, in an NMR magnet rather than in an organism.

It does not, however, make every nucleus eligible. A T_{1n} of order a second is measured for protons in proteins[34]—at high field, the caveat of the preceding paragraph—but ^{14}N is quadrupolar, and at the measured flavin quadrupole couplings and a protein rotational correlation time it relaxes in ~ 0.2–0.3 μs at the geomagnetic field (supplementary material, Sec. S12)—faster than any plausible turnover. Two of the three nuclei in our model are the flavin ^{14}N, and they are therefore wiped every cycle whatever the turnover rate. Restricting the register to the tryptophan proton leaves IPC = 2.78 against a floor of 2.00, an excess of 0.78 where the full register gave 2.42; the time-resolved kinetic readout, which needs all three nuclei, falls exactly to its floor (supplementary material, Sec. S9). The criterion survives on a proton register, at 32% of the excess capacity with the survivor register (29% with the product register) and through the CIDNP channel alone. The curve of Fig. 3(b) and every number derived from it above assume all three nuclei relax together; the register the biology can keep must be scanned as such. Wiping the two ^{14}N every cycle, carrying the product and depolarising the proton alone with probability q_H gives an eight-channel capacity of 1.85 at $q_\mathrm{H} = 0$ that holds to 1.89 at $q_\mathrm{H} = 0.5$ and then falls, through 1.67 at 0.8 and 1.30 at 0.9, to the floor 1.00 at $q_\mathrm{H} = 1$ (supplementary material, Sec. S9). Feeding the proton T_1 into *this* curve, as Sec. S12 does for the full one, caps the horizon at 2.6 ms on the intramolecular term and 1.7 ms with the bath, and moves the boundary from 8.3 to 4.0 ns and from 5.4 to 2.6 ns respectively: the register the biology can keep must reorient 3.8 (5.9) times faster than the protein, not 1.8 (2.8). That curve still hands on only the third of the molecules that recombined by τ. Closing the budget over the rest tightens it further (supplementary material, Sec. S9): if only the recombined branch is regenerated and the others are replaced, the cap is 1.0 (0.7) ms and the boundary 1.6 (1.0) ns, a factor of 10 (15); if both branches continue, their polarisations largely cancel and the band is not reached at any τ_c at the memory threshold used here. We regard this as the most consequential restriction in this work, because it replaces an assumed relaxation time with a measured one at the cost of most of the capacity.

This is the constraint that actually decides whether a radical pair can serve as a biological reservoir:

> *The same nuclear register must be re-used across successive turnovers, and successive turnovers must be driven by distinct input samples. The register must be carried by protons—the flavin ^{14}N is quadrupolar and*

cannot hold one—and must reorient with $\tau_c \lesssim 5.4$ ns, since at the geomagnetic field a proton whose dipolar vector reorients only with the protein relaxes in 2.4 ms on the intramolecular term alone, 1.6 ms with the intermolecular bath. Given all of that, the memory horizon is $(\mathrm{MC} - C_0) \times T_{\mathrm{turnover}}$, bounded above by $(\mathrm{MC} - C_0) \times T_{1n}$; without it, the system retains nothing.

This criterion is a statement about a spin register. We should be exact about how much of it needed the spin dynamics. The quantity $\mathrm{MC} - C_0 \simeq 1.9$, and with it the horizon, comes from the reservoir calculation. The constants that decide whether the horizon reaches the neural band divide unevenly. The proton T_1 of 2.4 ms and the photon budget are relaxation theory and photophysics and would have followed without the master equation. The 5.4 ms cap and the $\tau_c \lesssim 5.4$ ns requirement would not: both are obtained by feeding $q = 1 - \exp(-T_{\mathrm{d}}/T_{1n})$ into the measured $\mathrm{MC}(q)$ curve, which is itself an output of the master equation. The population-versus-coherence finding is real, but at the operating point it changes none of those numbers. We therefore claim the spin dynamics for the capacity, not for the criterion's parameters. We should not, however, overstate the distinction. Once q is parameterised as $q = 1 - \exp(-T_{\mathrm{d}}/T_{1n})$, the criterion's quantitative content is a comparison of two times, T_{turnover} against T_{1n}, just as the argument it replaces compared T_{turnover} against τ. What changes is not the form of the argument but which constant appears in it, and that constant is the longer one—by about 2×10^3 for a protein-bound proton at the geomagnetic field, and by $\sim 4 \times 10^5$ for a register that reorients with $\tau_c = 0.1$ ns. The genuinely non-temporal content is the binary chemical requirement that the pair be regenerated on the same molecule.

It is falsifiable by chemistry: it fails if the radical pair is regenerated on a fresh molecule each turnover, or if T_{1n} in the diamagnetic intermediate is faster than the turnover rate. We note that neither T_{turnover} nor T_{1n} has been measured in vivo for a neuronal cryptochrome, so as applied to neurons this is a conditional whose antecedents are not yet accessible. The in vitro route below is proposed because it makes both of them controllable.

H. The register is the product, and that helps

The results above propagate to the next turnover the nuclear state of the pair that did *not* recombine, renormalised. That is a defensible reservoir, but it is not the object the argument of this paper is about: our claim is that the register leaves in the diamagnetic product. The two are disjoint sub-populations of each turnover. We therefore reimplemented the cycle so that the state handed forward is the nuclear density matrix actually carried away by the recombined molecule,

$$\sigma_{\mathrm{nuc}} \propto \int_0^{\tau} \left[k_S \,\mathrm{Tr}_e(P_S \rho P_S) + k_T \,\mathrm{Tr}_e(P_T \rho P_T) \right] \mathrm{d}t, \quad (2)$$

with the electron pair reborn unpolarised, and recomputed every readout route.

TABLE VI. Capacity with the survivor register against the physically faithful product register.

readout route	ch.	survivor	product
end-point singlet yield	1	1.02	1.01
time-resolved kinetics	5	1.97	**2.77**
both exit channels, time-resolved	10	1.97	2.78
kinetics + CIDNP on the product	8	4.42	**4.96**

The chemically correct register is the better one. Time-resolved kinetics rise from 1.97 to 2.77 and the CIDNP route from 4.42 to 4.96; the linear memory capacity rises correspondingly, from 1.93 to 2.40 and from 2.82 to 3.12. The survivor-register numbers are therefore the conservative ones, and they are what we report throughout, giving Eq. (2) as the physically faithful variant. Every result that *reduces* the paper's claims—the proton-only register, the collapse of the kinetic readout under turnover jitter, and the anisotropy cost—has been re-run with the product register and survives (supplementary material, Sec. S4). The one quantitative difference is that the kinetic readout degrades more gracefully under half a cycle of jitter with the product register, retaining 18% of its excess at half a cycle where the survivor register retains under 1%; it reaches zero near one and a half cycles. No conclusion in this work depends on the choice.

I. What the numbers depend on

Three sensitivities deserve to be stated.

Discretisation and sampling. The linear memory capacity—the quantity the turnover-clock claim is about—converges in the time grid: MC = 2.790, 2.820, 2.821 for $n_t = 48$, 96, 192, a change of 0.03% across the last refinement, against the 8.2% effect being claimed. The five-channel capacity is less well converged (1.96, 1.97, 1.95) and we do not base claims on its third digit. A single input realisation cannot support four significant figures, so the turnover-clock scan uses twelve.

Regularisation. The linear readout is a ridge with $\lambda = 10^{-6}$ on unstandardised features, and this choice is load-bearing: the time-resolved channels are nested partial integrals of one signal, so the Gram matrix is ill-conditioned ($\kappa \approx 10^{10}$), and the recovered capacity runs from 4.50 at $\lambda = 10^{-12}$ to 1.93 at $\lambda = 10^{-6}$ to 1.00 at $\lambda = 10^{-1}$ for the kinetic route. Standardising the features raises the values substantially (to 3.69 at $\lambda = 10^{-6}$). The reported capacities are thus *conservative* with respect to this choice rather than inflated by

it, but they are not choice-independent, and we do not present them as though they were.

Counting statistics. The physically meaningful regulariser is the noise on a real product pool, and it must be counted as a pool is counted: a yield is the fraction of N molecules that recombined, and a product polarisation is the mean of Ny_S values of ±1, whose variance $(1-P^2)/Ny_S$ does not shrink with the polarisation. Sampling the eight channels that way (supplementary material, Sec. S7) leaves the CIDNP excess over its floor at 2.41 of its noiseless 2.42 for $N = 10^{10}$, 2.35 for 10^9, 1.93 for 10^8, 0.96 for 10^7 and 0.49 for 10^6. A relative Gaussian noise of size $N^{-1/2}$ per channel, the model one might first write down, understates this by about two orders of magnitude in N, because the polarisations are of order 10^{-2} and their counting noise is not proportional to them. The readout is therefore counting-noise limited below about 10^9 molecules per pool, and we do not know of a measurement placing the cryptochrome copy number of a single cell near that figure. This does not address the distinct question of whether 10^{10} copies driven by a common input but not phase-locked to a common turnover schedule retain the single-molecule capacity after pooling; that is computed in Sec. S10 and reported under Limitations below, where it turns out to be the more severe of the two effects.

III. DISCUSSION

Three expectations, and why they fail. Three claims that are natural to make about this system—that a chemical readout is too impoverished to be useful, that a spin reservoir of a few spins beats a classical system of the same size, and that microsecond radical-pair lifetimes put biological computation out of reach—all fail when the model is driven as a real turnover and compared symmetrically, though not in the way one might expect. The first fails only because a chemical readout need not be scalar: the endpoint yield is exactly memoryless, and what carries memory is the time course of the product and the polarisation it leaves behind. The quantum advantage is not there, and at the biologically relevant operating point no accounting we can construct puts the radical pair ahead by more than its own seed scatter. The timescale gap does not vanish but moves: the pair lifetime is the wrong clock, and the constant that replaces it is the nuclear relaxation time, which at the geomagnetic field is milliseconds rather than seconds. What remains is a chemical criterion, together with the structural fact that the register stores populations rather than coherence and is therefore governed by T_{1n}. That fact does not, however, rescue the biological reading: at the geomagnetic field extreme narrowing makes T_{1n} and T_{2n} equal, and both short, so what the criterion demands is a register that reorients fast enough to lengthen them.

Relation to prior work. Quantum reservoir computing in spin ensembles is established, including the use of dissipation as a resource and the exponential-state-space argument.[13,26–29] We do not claim these as new, and—having found no advantage under a symmetric comparison—we do not add to them. The question of whether nuclear polarisation accumulates usefully across successive cryptochrome photocycles has been asked before, by Wong *et al.*[35], who found that the gain available in an idealised model does not survive conditions approaching those in vivo. Our criterion is the chemical statement of what would have to be true for it to survive, and our conclusions are consistent with theirs: the reservoir works only in a narrow and specific corner of parameter space. Photo-CIDNP itself has been observed in a flavoprotein[40] though not yet in cryptochrome, and the underlying polarisation mechanism is standard[18]. Our contribution is to take a chemically realised radical pair, drive and read it as chemistry actually would, and convert a long-standing qualitative conjecture[6,7] into a criterion that experiment can test. The decoherence objection usually cited against such proposals[9] does not apply directly here: it evaluates superpositions of ion positions and tubulin dipole states, not nuclear spins, which is precisely why nuclear-spin proposals were advanced in reply to it. The objection that does apply is the one this work inherits—nuclear relaxation in the diamagnetic product—and the relevant history is discouraging. The long ^{31}P coherence times underpinning Ref.[6] are theoretical estimates rather than measurements, and have since been bounded at 37 min and expected at "a few seconds" by spin-dynamics calculation[36], while ^{31}P singlet order in pyrophosphate is measured to relax *faster* than T_1[33]. We therefore ground our own relaxation assumption on measured proton T_1 values in proteins rather than on estimate, and restrict the register accordingly (supplementary material, Sec. S9). The result also sharpens why CIDNP matters here: it is the readout that recombination itself writes, and under the idealised premise of a non-interacting pair it is the only mechanism by which information reaches the nuclei at all (Sec. II E).

The spin-dynamics literature on the cryptochrome pair has itself moved since the parameters used here were set, and three of its recent results constrain what the present model can and cannot say. Thermal motion of the protein modulates the hyperfine tensors on picosecond-to-microsecond timescales, and simulations that carry that modulation explicitly find both relaxation that curbs the compass[47] and frequency windows in which it helps[48]; our anisotropy and τ_c scans are static by comparison, so the register criterion is stated for a fixed tensor and would have to be re-evaluated against a time-dependent one. Decoherence itself has been shown to enhance the sensitivity of a radical-pair sensor under some conditions[49]; that concerns the yield's field dependence, not the capacity, and our finding that electronic dephasing over 50–2000 ns leaves the capacity within 2% is consistent with it rather than opposed. Tightly bound pairs with large exchange and dipolar couplings, long thought magnetically inert, retain magnetosensitivity through a quantum

Zeno mechanism[50], which is the physical reason the coupling premise of Sec. II E cannot be settled by assuming $J = 0$; the couplings themselves are now estimated for the FAD–TrpH pair together with the half-field parameter they produce[51]. And the five-spin reduction that limits every quantity here is no longer forced by cost: tensor-network propagation of open radical-pair dynamics has reached sixty spins[52], so the nuclear register could be modelled at its true size. On the readout side, the photo-CIDNP that our best route relies on is being measured in flavoproteins with increasing control[53,54], so the question whether the product's polarisation is readable is an experimental one with a growing literature, not a presumption.

Limitations. The model is a five-spin reduction. Two of its simplifications are relaxed here, with opposite outcomes. Treating ^{14}N correctly as spin-1 rather than spin-1/2 *increases* the excess capacity by 28%, so that approximation was conservative. Anisotropy is not: scanning an axial hyperfine tensor at fixed isotropic average costs two-fifths of the capacity with the survivor register and two-thirds with the product register by $\eta = 0.5$, and the capacity vanishes exactly at the Ising point where the perpendicular components go to zero, which identifies those components as what carries the memory (supplementary material, Sec. S11). The published tensor set of the pair[12,21], used on the three nuclei of this model with the field orientation scanned and averaged, and with a fourth nucleus added (supplementary material, Sec. S11), settles the direction: the full register's CIDNP excess falls from 2.96 to 0.93, because near-axial nitrogens are written far less than isotropic ones, while the proton-only register the criterion rests on retains 0.60 against 0.86 in the orientation average (0.98–1.47 with a second proton), and the kinetic route, dead on a proton register in the isotropic model, retains 0.74 there. The headline capacities are therefore the isotropic model's and overstate the full register; the criterion's numbers are not loosened. The tryptophan nuclei beyond Hβ1 and H1, and spin-1 nitrogen in the full model, are still not treated. The capacity estimator uses degree-2 targets, so higher-order nonlinear capacity is not counted. Whether a *neuronal* cryptochrome forms a separated radical pair at all is unestablished—CRY1/CRY2 are primarily circadian transcriptional regulators[15], and a behavioural test of magnetic-field effects in *Drosophila*, where cryptochrome is abundant, found none[55]—so the biological reading of our criterion is conditional. The spin-dynamics results do not depend on that condition.

Four further conditions bound the biological reading. Three we quantify here, and each narrows the claim.

(i) *Which nuclei hold the register.* Not all of them. Nuclear *dephasing* turns out to be irrelevant—the register is a population register—so the binding constant is T_{1n} as assumed. But T_{1n} is not one number. Protein-bound ^{1}H relaxes on the second timescale in the measurements available[34]—all of them at NMR field strengths, a caveat item (iv) below makes decisive—whereas ^{14}N is quadrupolar and, at the measured flavin quadrupole couplings and a protein correlation time, relaxes in ~ 0.2–0.3 μs at the geomagnetic field and in single microseconds even at 9.4 T[37,38]—faster than any physiological turnover at either. Two of our three nuclei are ^{14}N, and they cannot hold a register at any physiological turnover rate. Restricting the register to the proton leaves 29% of the excess capacity and destroys the time-resolved kinetic readout entirely. Of the two routes we tested under this restriction, CIDNP is the one that survives; the other five routes were not re-evaluated with a restricted register (supplementary material, Sec. S9). The turnover-clock argument survives, on measured rather than assumed relaxation data; the capacity does not survive intact.

(ii) *Ensemble averaging.* Pooling $\sim 10^{10}$ copies is safe against heterogeneity—of the couplings and, separately, of the relaxation rate—but not against desynchronisation. Letting τ_c vary log-normally over a sixteen-fold range, so that copies retain different fractions of their register, leaves the CIDNP excess at 1.96 against 1.95 for a uniform pool: a change of 0.7 per cent, within the seed scatter, and a mean-field control at the pool's own mean survival reproduces it to within 0.01, so a heterogeneous pool is, to this precision, a homogeneous one at its mean T_{1n} (supplementary material, Sec. S10). The kinetic readout's memory falls away with the spread of copy phases: with the product register its excess declines from 1.72 at zero jitter to 0.32 at half a cycle, 0.18 at one cycle and 0.02 at one and a half, and with the survivor register it is already at 0.01 by half a cycle. Two cautions attach to reading those numbers as a smooth decay. The survivor sequence is not monotonic—it returns to 0.11 at one cycle—and the register-wiped floor itself rises steeply beyond one cycle (from 1.01 to 1.86 for the product register), because pooling copies that sit whole cycles apart is itself a delay line. The excess is the right quantity precisely because it subtracts that, but the underlying capacities do not fall monotonically and we do not claim they do. CIDNP retains 40–49% of its excess even at two cycles of jitter, depending on the register, because it integrates over the cycle rather than sampling it (Sec. S10). Synchronised turnover is thus a requirement of the kinetic readout specifically. Both restrictions—nuclide and desynchronisation—leave CIDNP as the surviving one of the two routes tested under each.

(iii) *Readout signal-to-noise.* The ridge parameter is an implicit noise model and the recovered capacity depends on it (Sec. II I); the reported values are conservative with respect to that choice. Counting noise is not: a pool of 10^8 molecules retains 80% of the full register's excess and 63% of the proton register's, and a pool of 10^6 retains essentially none of either (Sec. II I; supplementary material, Sec. S7). The criterion therefore carries a third constant besides T_{turnover} and T_{1n}: the number of molecules read as one pool.

(iv) *Falsifiability.* Neither T_{turnover} nor T_{1n} has been measured for a neuronal cryptochrome, but both can be predicted from measured constants, and doing so sharp-

ens the criterion into something testable without a neuron (supplementary material, Sec. S12). Two results follow, and the first is adverse. Nuclear relaxation at the geomagnetic field is in the extreme-narrowing regime, where $1/T_1 \propto \tau_c$ is maximal, so the second-long T_{1n} we assumed is a *high-field* value: a proton on a 60 kDa protein relaxes in 2.4 ms at 50 μT against 9.0 s at 9.4 T. What sets the ceiling is not whether the register is chemically bound to the protein but the effective correlation time of its dipolar vector, $\tau_{\text{eff}} = S^2\tau_c + (1 - S^2)\tau_{\text{int}}$, together with the dipolar sum $\Sigma = \sum_j S_j^2 r_j^{-6}$ over intramolecular *and* intermolecular partners. For a vector that reorients only with the intact 60 kDa protein ($S^2 = 1$, $\tau_c = 15.2$ ns) the intramolecular geminal term alone gives $T_1 \leq 2.4$ ms and a ceiling of ≤ 5.4 ms; adding the intermolecular proton bath—whose size we calibrate on the water row of our own validation table, at 54% of the intramolecular rate—shortens these to 1.6 and 3.5 ms. That row is an empirical anchor rather than an independent check: f is defined as its residual, so the agreement it then shows is an identity. Fast internal motion lengthens T_1: the intramethyl H–H vector of the flavin 8α-CH_3 has the exact geometric value $S^2 = 1/4$, because that vector does *not* co-rotate rigidly with the protein even though the methyl is covalently attached, and the intramethyl coupling alone would give $T_1 = 5.0$ ms and a ceiling of 11.2 ms—inside the band. That escape survives neither the bath, which moves the methyl's own boundary from 17.1 to 9.6 ns, nor the restriction to the register the biology can keep: on the proton-only product register the methyl's ceiling is 5.4 ms without the bath and 3.0 ms with it, below the band either way. The criterion is therefore quantitative and vector-specific rather than a statement about binding: the register's dipolar vector must lose orientational memory four to six times faster than the intact protein ($\tau_{\text{eff}} \lesssim 2.6$–$4.0$ ns for Trp Hβ, 4.6–8.2 ns for the methyl), factors that would be 1.8–2.8 if all three nuclei could be kept. Second, the drive cannot be light: with the measured flavin extinction coefficient, a single flavin turns over once per 7.7 s in full sunlight and once per some hundreds of years inside a skull, so a photochemical cycle fails by about ten orders of magnitude and the regeneration must be catalytic. We note what that requirement does to the model: the Hamiltonian is that of the photogenerated [$FAD^{\bullet-}$ $TrpH^{\bullet+}$] pair, and no dark, catalytic route that regenerates *that* pair on a cryptochrome is established; the dark reoxidation of the flavin proceeds through a different pair. The criterion therefore names a requirement on whatever flavoenzyme cycle is proposed to meet it, not a cycle known to exist. The criterion that survives is correspondingly specific—the pair regenerated catalytically on a register with $\tau_c \lesssim 2.6$–4.0 ns (5.4 ns if all three nuclei could be kept), at a turnover interval that depends on how mobile that register is—10–14 ms at $\tau_c = 5$ ns, widening to 7.3–79 ms at 1 ns and 5.6–790 ms at 0.1 ns, with the intermolecular bath included—and both of its parameters are measurable in vitro by relaxation dispersion and steady-state kinetics.

(v) *What carries the input.* The input of this paper is the singlet character s of the pair at birth, swept over $[0, 1]$, and no chemistry is named that modulates it over that range. Three carriers that chemistry or physics could modulate were therefore driven through the same reservoir with singlet birth throughout, each against its own memoryless floor (supplementary material, Sec. S3): the field magnitude over 25–75 μT, the singlet recombination rate over 0.5–1.5 μs^{-1}, and a small-signal s. The field is the best carrier we found—CIDNP-route excess 3.14 against 2.96 for the full-range s encoding—and the rate carrier retains 2.42, so the criterion does not depend on a chemically unmotivated input. It does depend on the input's amplitude: at a tenth of the range the excess is 1.75, at a hundredth 0.76, and at a thousandth nothing survives the readout's regulariser. A magnetoreceptor's natural input, the field, is also the one that cryptochrome is equipped to receive; whether a neuron modulates it is a separate question.

What would settle it. The criterion is addressable in vitro before it is addressable in a neuron. Covalently linked donor–bridge–acceptor radical pairs have synthetically tunable exchange coupling and lifetime and are read by transient EPR and optical/CIDNP spectroscopy.[2,14,41] Repeatedly cycling such a system and asking whether the product yield of turnover k depends on the input at turnover $k-1$, $k-2$, ... measures the memory horizon directly, and varying the interval between cycles tests the register-reuse criterion where it is sharpest.

IV. CONCLUSION

Whether a radical pair can compute reduces, once the chemistry is left in, to the register-reuse criterion: two constants, the proton T_1 of the diamagnetic product at the geomagnetic field and the turnover interval, both measurable in vitro. Driven and read as a chemical turnover rather than as an abstract quantum channel, a cryptochrome-type radical pair is a reservoir whose output ordinary chemistry can recover: time-resolved product kinetics give IPC = 2.0 against a floor of 1.0, and the nuclear polarisation carried off by the product reaches 4.4 against a floor of 2.0, with no spectroscopic access required. Under a specific matched-hopping incoherent limit 82% of that capacity is lost, and the loss is attributable to the electrons rather than to erasure of the nuclear register—and yet there is no quantum advantage: at the biologically relevant operating point a classical network matched channel-for-channel exceeds the radical pair. The obstacle usually invoked, a microsecond memory horizon, does not survive scrutiny: the register is nuclear and survives recombination, so the clock is the turnover interval, and a 10–100 ms turnover places the horizon at 19–188 ms—so long as the register itself lasts that long, which for a vector reorienting only

with the whole protein nucleus at the geomagnetic field it does not. What decides the question instead is whether the same nuclear register is reused from one turnover to the next; if it is not, the capacity falls to a memoryless $\mathrm{MC} = 1$. Because that register stores its information in nuclear populations rather than in nuclear coherence, the constant that bounds it is a nuclear T_1. That distinction buys nothing at the operating point: the geomagnetic field puts the register in extreme narrowing, where T_{1n} and T_{2n} are equal. It cannot, however, be any nucleus. The flavin ^{14}N is quadrupolar and relaxes in microseconds, so the register must be carried by protons; that restriction removes seven tenths of the *excess* capacity—the IPC itself falls by two fifths—and leaves the product's nuclear polarisation as the only readout we tested that still works, a conclusion reached independently by asking what survives when the copies in a 10^{10}-molecule pool fall out of step. We offer the criterion as the thing to test, in the sharpened form that the relevant constant is the proton T_1 of the diamagnetic product *at the geomagnetic field*, which our own calculation puts at 2.4 ms—1.6 ms once the intermolecular bath is restored—for a nucleus whose dipolar vector reorients only with the protein, rather than at the second suggested by high-field measurements—so that the register must reorient four to six times faster than the protein ($\tau_{\mathrm{eff}} \lesssim 2.6$–$4.0$ ns) for the horizon to reach the neural band at all. The turnover interval of a neuronal cryptochrome has not been measured.

SUPPLEMENTARY MATERIAL

See the supplementary material for S1, liouville-space solver and the coherent limit; S2, capacity estimator and its controls; S3, the injection map; S4, chemically accessible readout routes; S5, coherence fraction at matched hopping rate; S6, classical baselines; S7, turnover clock and the register-reuse criterion; S8, a semiclassical reference for the coherence fraction; S9, which nuclei can hold the register; S10, ensemble averaging; S11, anisotropic hyperfine coupling and the nuclear spin quantum number; S12, predicting the two unmeasured constants; S13, spin parameters; S14, dependence of the cryptochrome point on the electron–electron coupling.

ACKNOWLEDGMENTS

Generative AI tools were used in this work as described in Appendix B: Claude Opus 5, Claude Opus 5.2 and Claude Fable 5.1 (Anthropic) through Claude Code, and GPT-5.6 (OpenAI) through Codex CLI. All numerical results are reproducible from the open-source `qbscreen` package.

AUTHOR DECLARATIONS

Conflict of interest

The authors have no conflicts to disclose.

Author contributions

H.W. conceived the study, developed the master-equation and reservoir code, ran the simulations, and wrote the manuscript. T.T. contributed to the spin-dynamics formulation and the analysis of the nuclear-register criterion. Both authors reviewed the results and approved the manuscript.

DATA AVAILABILITY

All analyses are implemented in the open `qbscreen` package, `https://github.com/deeptell-inc/new_Q_brain`. The correlated-pair-birth injection is in `qbscreen/corrected_injection.py`; the chemically accessible readout routes, the turnover-clock scan and the register-reuse criterion are in `qbscreen/readout_routes.py`; the protocol-consistent capacity, coherence fraction and classical baselines are in `qbscreen/final_numbers.py`; the master-equation solver is in `qbscreen/master_equation.py`. Numerical outputs are in `simulation_results/`. The capacity bound and the agreement of the solver with the closed-form coherent limit are checked in `qbscreen/tests/` (316 tests, all passing). Thirty-seven tests in `test_panel_claims.py` bind the headline quantities and load-bearing claims, and a further one hundred and forty-four in `test_table_rows.py` bind the capacity and standard-deviation cells of every table in the main text and the supplementary material, with the expected values transcribed from the manuscript rather than from the data files. The opposite direction is checked by `test_printed_cells.py`, which parses the numeric cells back out of the `.tex`: the headline tables are matched column by column against their data files, and every other printed table cell must equal some shipped value. The suite therefore fails if either side moves. The spin-parameter table (S19) is now bound as well, by `test_spin_parameters.py`: it parses the printed values and compares them to the dictionaries the Hamiltonian is built from, checking the millitesla column as the megahertz column divided by the electron gyromagnetic ratio rather than as an independent input. Derived-time and raw-IPC columns remain unbound. Hand-typed cross-document references to supplement tables, which LaTeX cannot check across separate documents, are bound against `supplementary.aux` by `test_cross_document_refs.py`. This closes the kind of drift in which a figure quoted from a five-channel column is printed beside an eight-channel table. The LaTeX sources these tests read accompany the submission with the freeze manifest and are not distributed in the public repository; on a clone those tests skip rather than fail.

Appendix A: Model and methods

1. Reservoir Hamiltonian

We model the reservoir as $N = 5$ spins—two radical-pair electrons (e_1, e_2) and three spin-1/2 nuclei—with Hamiltonian (in MHz). At the cryptochrome point the three nuclei carry the hyperfine couplings of ^{14}N5 and ^{14}N10 on FAD$^{\bullet-}$ and Hβ1 on TrpH$^{\bullet+}$ (supplementary material, Table S29), with ^{14}N modelled as spin-1/2; no ^{31}P is present in the cryptochrome model, and the ^{31}P-based proposal of Ref.[6] is therefore *not* the system tested here.

$$H = \omega_e \left(S_z^{e_1} + S_z^{e_2}\right) + \sum_{\mu=x,y,z} \Big[A_P\, S_\mu^{e_1} I_\mu^{P} + A_{H_1} S_\mu^{e_1} I_\mu^{H_1} + A_{e_2} S_\mu^{e_2} I_\mu^{H_2} + J\, S_\mu^{e_1} S_\mu^{e_2} \Big], \tag{A1}$$

where $\omega_e = g_e \mu_B B/h$ is the electron Zeeman frequency. Each radical carries its own nuclei—they are not shared, which is what a spatially separated pair means—and the exchange J links the two electrons. At the engineered operating point we use $A_{e_1 a} = 80$, $A_{e_1 b} = 40$, $A_{e_2 a} = 15$ MHz, $J = 2$ MHz and $B = 1$ mT: representative molecular-spin-RC values chosen to give effective hyperfine writing, not a fit to a specific protein. For the cryptochrome point we set $B = 50$ μT and $\tau = 1$ μs, with hyperfine magnitudes representative of the dominant FAD$^{\bullet-}$ and TrpH$^{\bullet+}$ couplings[21,22], and take $J = 0$ with no dipolar term as the reference premise: the two radicals interact only through recombination. That is an idealisation of "separated". The source for the separated pair[23] reports, for FAD–W324 at $r = 1.90$ nm, a point-dipole coupling $D = -11.4$ MHz and an exchange coupling whose magnitude lies between 1.3 and 126 MHz with undetermined sign, the two partially cancelling near 2 nm. Section II E and supplementary material Sec. S14 rerun every cryptochrome-point quantity with those values in Eq. (A1) (D entered as the axial tensor $2D(\boldsymbol{S}_1\cdot\hat{n})(\boldsymbol{S}_2\cdot\hat{n}) - \frac{2D}{3}\boldsymbol{S}_1\cdot\boldsymbol{S}_2$, both signs of J, inter-radical axis parallel and perpendicular to B) so that what depends on the reference premise is stated as a number rather than assumed away. The capacity is not fine-tuned to these numbers: at the engineered point it varies by under 2% over $T_2^e = 50$–2000 ns, and that insensitivity is estimator-robust (a full-rank readout gives -1.7% over the same range). Below the operating point the two estimators disagree: the ridge readout recovers a steep fall (-47% from 50 to 5 ns), but this is an artifact of the fixed absolute ridge penalising the small-amplitude features at short T_2^e — a full-rank readout gives $+2.5$% for the same comparison (cf. the Regularisation paragraph of Sec. II I). The insensitivity claimed here is therefore to T_2^e at or above τ, and we make no claim below the operating point. The register criterion of Sec. II G, however, is *not* independent of the hyperfine assignment—it is carried by the perpendicular tensor components and vanishes with them (supplementary material, Sec. S11).

2. Master equation and propagator

Each reservoir update is the completely positive, trace-preserving map $\rho \mapsto e^{\mathcal{L}\tau}\rho$ generated by the Haberkorn master equation[30] (whose spin-selective recombination operator we adopt in its standard form[17]) with electronic

pure dephasing,

$$\frac{d\rho}{dt} = -i[H,\rho] - \tfrac{1}{2}\{k_S P_S + k_T P_T, \rho\} + \sum_i \frac{2}{T_2^e}\left(S_z^i \rho S_z^i - \tfrac{1}{4}\rho\right), \tag{A2}$$

solved exactly in Liouville space. Couplings are quoted in MHz throughout; H is multiplied by 2π before propagation, so that $e^{\mathcal{L}\tau}$ acts in angular frequency (rad μs^{-1}) with τ in μs, and the rates k_S, k_T and $2/T_2^e$ are in μs^{-1}. The convention is pinned to physics rather than to internal consistency: a lone electron propagated by the same code at 1 mT precesses at $g_e\mu_B B/h = 28.0$ MHz (`test_units.py`). The coefficient $2/T_2^e$ makes the single-electron transverse coherence decay as e^{-t/T_2^e}, so T_2^e is the single-electron transverse relaxation time in the usual sense. The third term is electronic decoherence; it is not what supplies the fading memory here, which comes from the openness of the chemical cycle and from inter-cycle nuclear depolarisation. We verified that, in the limit $1/T_2^e \to 0$ and $k_S = k_T$, the Liouville-space solver reproduces the closed-form coherent Green's-function singlet yield to better than 10^{-9} (supplementary material). Whether recombination acts within the update step depends on which result is being computed; the two conventions are set out under "Reservoir protocol" below.

3. Reservoir protocol

We adapt the temporal-QRC protocol of Ref.[26], but the drive is *not* the single-electron reset used there. At step k the nuclei are retained and the electron pair is recreated as a spin-correlated pair whose singlet character encodes the input,

$$\rho_e(s_k) = s_k\,|S\rangle\langle S| + (1-s_k)\,P_T/3, \tag{A3}$$

so that the reservoir state entering cycle k is $\rho_e(s_k)\otimes\sigma_{\mathrm{nuc}}^{(k)}$. The state then evolves for the pair lifetime τ under $e^{\mathcal{L}\tau}$; a set of observables is read out, and a linear (ridge) readout is trained on them. Inputs s_k are i.i.d. uniform on $[0,1]$; the first 80–100 steps are discarded as washout. The washout discards initial transients, so the reported capacities are independent of the initial spin state—a consequence of the fading-memory (echo-state) property.

Two operating points are used and must not be conflated. The *engineered* point ($J = 2$ MHz, $B = 1$ mT, $\tau = T_2^e = 50$ ns) is a molecular-spin-RC reference and is where the twelve-observable capacity, the coherence fraction and the classical baselines of Secs. II are evaluated; it lies outside the weak-field separated-pair regime that motivates the biological reading, and we flag every number taken there. The *cryptochrome* point ($J = 0$, $B = 50$ μT, $\tau = 1$ μs, $T_2^e = 1$ μs) is where the chemically accessible readouts, the turnover clock and the register criterion are evaluated. Recombination is likewise not uniform across the paper: the twelve-observable capacity is computed with $k_S = k_T = 0$ within the update step, whereas every chemical-readout, clock and criterion result is computed with spin-selective recombination acting throughout the cycle ($k_S = 1$, $k_T = 0.2$ μs^{-1}).

The single-electron reset of Ref.[26], in which only the input electron is replaced by a product state while the reduced state of the remainder is retained, is used in this work *only* as a control. It is not a chemical process, and for a separated pair ($J = 0$) it destroys the electron–electron correlation every cycle: since $\langle P_S\rangle = \frac{1}{4} - \langle \boldsymbol{S}_1\cdot\boldsymbol{S}_2\rangle$ with $\langle \boldsymbol{S}_2\rangle = 0$, the singlet probability is pinned at exactly 1/4 independently of the input, and the measured capacity is identically zero (supplementary material, Sec. S3). All results reported below use Eq. (A3).

4. Capacity measures

We quantify the reservoir with two information-theoretic benchmarks. The linear *memory capacity*[31] $\mathrm{MC} = \sum_{d\geq 0} r^2(s_{k-d})$ sums the squared multiple correlation for reconstructing delayed inputs. The *information processing capacity*[32] adds nonlinear functions of the input history, $\mathrm{IPC} = \mathrm{MC} + \sum_{\mathrm{nonlinear}} C$, evaluated here through degree-2 Legendre targets (diagonal $P_2(s_{k-d})$ and cross $s_{k-d_1}s_{k-d_2}$ terms). A fundamental theorem[32] bounds the total capacity by the number of linearly independent readout observables, $\mathrm{IPC} \leq n_{\mathrm{obs}}$; we use this bound throughout as a correctness anchor. It is an asymptotic bound for an orthonormal target family and is violated by a few per cent in finite samples, so satisfying it does *not* demonstrate that the readout observables are independent, and we make no such argument anywhere (supplementary material, Sec. S2).

All results are reproducible from the open `qbscreen` package; the module list is given under Data and code availability below. We note there that `ensemble.py`, `quantum_vs_classical.py` and the `run_reservoir` driver in `reservoir.py` implement the single-electron-reset control and are retained only as such, so no result reported here is taken from them; the capacity estimators in `reservoir.py` are used throughout (316/316 tests pass).

Appendix B: Use of generative AI tools

Generative AI tools were used in producing this work, and we state here how, to the standard that the use is reproducible rather than merely declared. *Code.* The `qbscreen` package was written in Claude Code sessions (Anthropic; Claude Opus 5 and 5.2, and Claude Fable 5.1) directed by H.W., who specified the physics, the protocols and the quantities to compute and reviewed every module. The Liouville-space solver, the injection map, the readout routes and the capacity estimators were implemented this way; their correctness rests not

on the authors' or the tools' assertion but on the tests listed under Data and code availability, which bind every printed number to the data files in both directions. *Text.* The manuscript, the supplementary material and the submission documents were drafted and revised with the same tools from the authors' outlines, results and instructions; every sentence was read and approved by the authors. *Adversarial review.* Before submission the analysis was put through a review protocol in which two model families—OpenAI Codex (GPT-5.6) and Anthropic Claude (Opus 5)—were given the manuscript, the code and the data independently and instructed to act as referees: to score the work, to list every claim they could falsify, and to refute by reproduction rather than by assertion (re-running the generators, recomputing printed cells, corrupting a value to see whether any test caught it); each then critiqued the other's report. A facilitating Claude session collected the reports and proposed resolutions, and the authors adjudicated every disagreement. Representative prompts were: "As a referee, score this manuscript on novelty, data quality and logic, and list every claim you can falsify by running the code"; "Reproduce Table 3 from the shipped JSON and report any cell that disagrees"; "Change one printed cell and report whether any test fails". The controls of supplementary material, Secs. S4–S12 and the test modules named under Data and code availability are the product of nine such rounds; several of them bound the claims more tightly than the uncontrolled calculation would, and the tighter bounds are what is reported. The full prompts, the panel reports and the authors' adjudications are kept in the manuscript's review log and are available from the authors. No AI tool is an author, and the authors take full responsibility for the content.

[1] U. E. Steiner and T. Ulrich, *Magnetic field effects in chemical kinetics and related phenomena*, Chem. Rev. **89**, 51 (1989).

[2] P. J. Hore and H. Mouritsen, *The radical-pair mechanism of magnetoreception*, Annu. Rev. Biophys. **45**, 299 (2016).

[3] T. Ritz, S. Adem, and K. Schulten, *A model for photoreceptor-based magnetoreception in birds*, Biophys. J. **78**, 707 (2000).

[4] K. Maeda *et al.*, *Magnetically sensitive light-induced reactions in cryptochrome are consistent with its proposed role as a magnetoreceptor*, Proc. Natl. Acad. Sci. USA **109**, 4774 (2012).

[5] J. Xu *et al.*, *Magnetic sensitivity of cryptochrome 4 from a migratory songbird*, Nature **594**, 535 (2021).

[6] M. P. A. Fisher, *Quantum cognition: The possibility of processing with nuclear spins in the brain*, Ann. Phys. **362**, 593 (2015).

[7] C. M. Kerskens and D. López Pérez, *Experimental indications of non-classical brain functions*, J. Phys. Commun. **6**, 105001 (2022).

[8] G. Buzsáki and A. Draguhn, *Neuronal oscillations in cortical networks*, Science **304**, 1926 (2004).

[9] M. Tegmark, *Importance of quantum decoherence in brain processes*, Phys. Rev. E **61**, 4194 (2000).

[10] N. Lambert, Y.-N. Chen, Y.-C. Cheng, C.-M. Li, G.-Y. Chen, and F. Nori, *Quantum biology*, Nat. Phys. **9**, 10 (2013).

[11] J. Cao *et al.*, *Quantum biology revisited*, Sci. Adv. **6**, eaaz4888 (2020).

[12] H. G. Hiscock *et al.*, *The quantum needle of the avian magnetic compass*, Proc. Natl. Acad. Sci. USA **113**, 4634 (2016).

[13] G. Tanaka *et al.*, *Recent advances in physical reservoir computing: A review*, Neural Netw. **115**, 100 (2019).

[14] M. R. Wasielewski *et al.*, *Exploiting chemistry and molecular systems for quantum information science*, Nat. Rev. Chem. **4**, 490 (2020).

[15] C. L. Partch, C. B. Green, and J. S. Takahashi, *Molecular architecture of the mammalian circadian clock*, Trends Cell Biol. **24**, 90 (2014).

[16] C. R. Timmel, U. Till, B. Brocklehurst, K. A. McLauchlan, and P. J. Hore, *Effects of weak magnetic fields on free radical recombination reactions*, Mol. Phys. **95**, 71 (1998).

[17] J. A. Jones and P. J. Hore, *Spin-selective reactions of radical pairs act as quantum measurements*, Chem. Phys. Lett. **488**, 90 (2010).

[18] P. J. Hore and R. W. Broadhurst, *Photo-CIDNP of biopolymers*, Prog. Nucl. Magn. Reson. Spectrosc. **25**, 345 (1993).

[19] H. Wakaura, *Systematic screening of brain neurotransmitter enzymes for radical-pair mechanism competence identifies candidates for magnetic field sensitivity*, Research Square preprint, doi:10.21203/rs.3.rs-9278975/v1 (2025).

[20] H. Wakaura, *Motional narrowing rescues entanglement in biological radical-pair systems: quantum error correction and reservoir computing beyond the breaking threshold*, Research Square preprint, doi:10.21203/rs.3.rs-9493052/v1 (2026).

[21] A. A. Lee, J. C. S. Lau, H. J. Hogben, T. Biskup, D. R. Kattnig, and P. J. Hore, *Alternative radical pairs for cryptochrome-based magnetoreception*, J. R. Soc. Interface **11**, 20131063 (2014).

[22] E. Schleicher *et al.*, *Selective* ^{13}C *labelling reveals the electronic structure of flavocoenzyme radicals*, Sci. Rep. **11**, 18395 (2021).

[23] O. Efimova and P. J. Hore, *Role of exchange and dipolar interactions in the radical pair model of the avian magnetic compass*, Biophys. J. **94**, 1565 (2008).

[24] W. Maass, T. Natschläger, and H. Markram, *Real-time computing without stable states: A new framework for neural computation based on perturbations*, Neural Comput. **14**, 2531 (2002).

[25] H. Jaeger and H. Haas, *Harnessing nonlinearity: Predicting chaotic systems and saving energy in wireless communication*, Science **304**, 78 (2004).

[26] K. Fujii and K. Nakajima, *Harnessing disordered-ensemble quantum dynamics for machine learning*, Phys. Rev. Applied **8**, 024030 (2017).

[27] K. Nakajima *et al.*, *Boosting computational power through spatial multiplexing in quantum reservoir computing*, Phys. Rev. Applied **11**, 034021 (2019).

[28] R. Martínez-Peña *et al.*, *Dynamical phase transitions in quantum reservoir computing*, Phys. Rev. Lett. **127**, 100502 (2021).

[29] P. Mujal, R. Martínez-Peña, J. Nokkala, J. García-Beni, G. L. Giorgi, M. C. Soriano, and R. Zambrini, *Opportunities in quantum reservoir computing and extreme learning machines*, Adv. Quantum Technol. **4**, 2100027 (2021).

[30] R. Haberkorn, *Density matrix description of spin-selective radical pair reactions*, Mol. Phys. **32**, 1491 (1976).

[31] H. Jaeger, *Short term memory in echo state networks*, GMD Report 152, German National Research Center for Information Technology (2002).

[32] J. Dambre, D. Verstraeten, B. Schrauwen, and S. Massar, *Information processing capacity of dynamical systems*, Sci. Rep. **2**, 514 (2012).

[33] D. E. Korenchan, J. Lu, M. H. Levitt, and A. Jerschow, ^{31}P *nuclear spin singlet lifetimes in a system with switchable magnetic inequivalence: experiment and simulation*, Phys. Chem. Chem. Phys. **23**, 19465 (2021).

[34] N. Tjandra, S. E. Feller, R. W. Pastor, and A. Bax, *Rotational diffusion anisotropy of human ubiquitin from* ^{15}N *NMR relaxation*, J. Am. Chem. Soc. **117**, 12562 (1995).

[35] S. Y. Wong, I. A. Solov'yov, P. J. Hore, and D. R. Kattnig, *Nuclear polarization effects in cryptochrome-based magnetoreception*, J. Chem. Phys. **154**, 035102 (2021).

[36] T. C. Player and P. J. Hore, *Posner qubits: spin dynamics of entangled* $Ca_9(PO_4)_6$ *molecules and their role in neural processing*, J. R. Soc. Interface **15**, 20180494 (2018).

[37]Y. A. Goddard, J.-P. Korb, and R. G. Bryant, *Water molecule contributions to proton spin–lattice relaxation in rotationally immobilized proteins*, J. Magn. Reson. **199**, 68 (2009).
[38]E. P. Sunde and B. Halle, *Mechanism of ^{1}H–^{14}N cross-relaxation in immobilized proteins*, J. Magn. Reson. **203**, 257 (2010).
[39]W. Bogner *et al.*, *Assessment of ^{31}P relaxation times in the human calf muscle: a comparison between 3 T and 7 T in vivo*, Magn. Reson. Med. **62**, 574 (2009).
[40]S. S. Thamarath, J. Heberle, P. J. Hore, T. Kottke, and J. Matysik, *Solid-state photo-CIDNP effect observed in phototropin LOV1-C57S by ^{13}C magic-angle spinning NMR spectroscopy*, J. Am. Chem. Soc. **132**, 15542 (2010).
[41]M. R. Wasielewski, *Energy, charge, and spin transport in molecules and self-assembled nanostructures inspired by photosynthesis*, J. Org. Chem. **71**, 5051 (2006).
[42]A. Sannia, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, *Dissipation as a resource for quantum reservoir computing*, Quantum **8**, 1291 (2024).
[43]F. Hu, S. A. Khan, N. T. Bronn, G. Angelatos, G. E. Rowlands, G. J. Ribeill, and H. E. Türeci, *Overcoming the coherence time barrier in quantum machine learning on temporal data*, Nat. Commun. **15**, 7491 (2024).
[44]K. Kobayashi, K. Fujii, and N. Yamamoto, *Feedback-driven quantum reservoir computing for time-series analysis*, PRX Quantum **5**, 040325 (2024).
[45]Y. Kora, H. Zadeh-Haghighi, T. C. Stewart, K. Heshami, and C. Simon, *Frequency- and dissipation-dependent entanglement advantage in spin-network quantum reservoir computing*, Phys. Rev. A **110**, 042416 (2024).
[46]Y. Hou *et al.*, *High-accuracy temporal prediction via experimental quantum reservoir computing in correlated spins*, Phys. Rev. Lett. **136**, 120602 (2026).
[47]M. Grüning *et al.*, *The effect of spin relaxation on magnetic compass sensitivity in ErCry4a*, ChemPhysChem **25**, e202400129 (2024).
[48]P. L. Benjamin, L. Gerhards, I. A. Solov'yov, and P. J. Hore, *Magnetosensitivity of model flavin–tryptophan radical pairs in a dynamic protein environment*, J. Phys. Chem. B **129**, 5937 (2025).
[49]J. Luo, *Sensitivity enhancement of radical-pair magnetoreceptors as a result of spin decoherence*, J. Chem. Phys. **160** (2024), doi:10.1063/5.0182172.
[50]J. R. Denton *et al.*, *Magnetosensitivity of tightly bound radical pairs in cryptochrome is enabled by the quantum Zeno effect*, Nat. Commun. **15**, 10823 (2024).
[51]S. Y. Wong, P. Benjamin, and P. J. Hore, *Magnetic field effects on radical pair reactions: estimation of $B_{1/2}$ for flavin–tryptophan radical pairs in cryptochromes*, Phys. Chem. Chem. Phys. **25**, 975 (2023).
[52]K. Hino, D. S. Frantzov, Y. Kurashige, and L. M. Antill, *Introduction to modeling radical pair quantum spin dynamics with tensor networks*, J. Chem. Phys. **164** (2026), doi:10.1063/5.0314129.
[53]J. Matysik *et al.*, *Spin dynamics of flavoproteins*, Int. J. Mol. Sci. **24**, 8218 (2023).
[54]A. Schmidt, H. Schneider, B. Illarionov, A. Bacher, M. Fischer, and S. Weber, *Photo-CIDNP of solvent-exposed flavins in flavoproteins*, J. Phys. Chem. Lett. **16**, 7058 (2025).
[55]M. Bassetto *et al.*, *No evidence for magnetic field effects on the behaviour of Drosophila*, Nature **620**, 595 (2023).